Nerve Theorem

Marcin Giembicki

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1 Introduction

In this semester project, I will state and prove the basic Nerve Theorem for Compact Closed Covers. My work is fully based on the following papers:

1. https://arxiv.org/pdf/2203.03571https://arxiv.org/pdf/2203.03571 [1]
2. https://mediatum.ub.tum.de/doc/1730967/1730967.pdfhttps://mediatum.ub.tum.de/doc/1730967/1730967.pdf [2]

I added small expansions on some of the details of proofs that were omitted in the papers. The nerve theorem presented will be for compact covers, as in [2], because the proof will be shorter than the one for closed covers, presented in [1]. Most of the time we work with compact sets anyway.

All illustrations and pictures were created by me.

2 What is nerve Theorem

2.1 Nerve complex of a covered space

Suppose we have a covered space consisting of:

$$X \text{ – a topological space in Euclidean space } \mathbb{R}^d,$$
$$\mathcal{C} = (C_i)_{i\in I} \text{ – covering of the space}$$
$$I \text{ – indexing set of the covering}$$

such that

$$\bigcup_{i\in I} C_i = X.$$

The *nerve complex* of a covered space $(X, \mathcal{C} = (C_i)_{i\in I})$ is the abstract simplicial complex $N(\mathcal{C})$ whose simplices are finite subsets of I such that the intersection of the corresponding sets is non-empty:

$$N(\mathcal{C}) = \big\{ J \subseteq I \;\big|\; \bigcap_{j\in J} C_j \neq \emptyset \;\wedge\; |J| < \infty \big\}.$$

2.2 Nerve Theorem for finite compact convex covers

The Nerve Theorem featured in this project states that if $(X, \mathcal{C} = (C_i)_{i\in I})$ is a covered space where $|I|$ is finite, $X \subseteq \mathbb{R}^d$, and all C_i for $i \in I$ are compact as well as convex, then X is homotopy equivalent to the geometric realization of $N(\mathcal{C})$.

3 Proof

Let SdN denote the barycentric subdivision of the nerve complex, and let $|A|$ be the geometric realization of an abstract simplicial complex A. For a given covered space $(X, \mathcal{C})$, one can construct two continuous functions,

$$\Phi\colon X \to |SdN(\mathcal{C})|$$
$$\Gamma\colon |SdN(\mathcal{C})| \to X.$$

We can then show that the composition $\Phi \circ \Gamma$ is homotopic to the identity map on $|SdN(\mathcal{C})|$, denoted by $id_{|SdN(\mathcal{C})|}$, and that $\Gamma \circ \Phi$ is homotopic to the identity map on X, denoted by id_X.

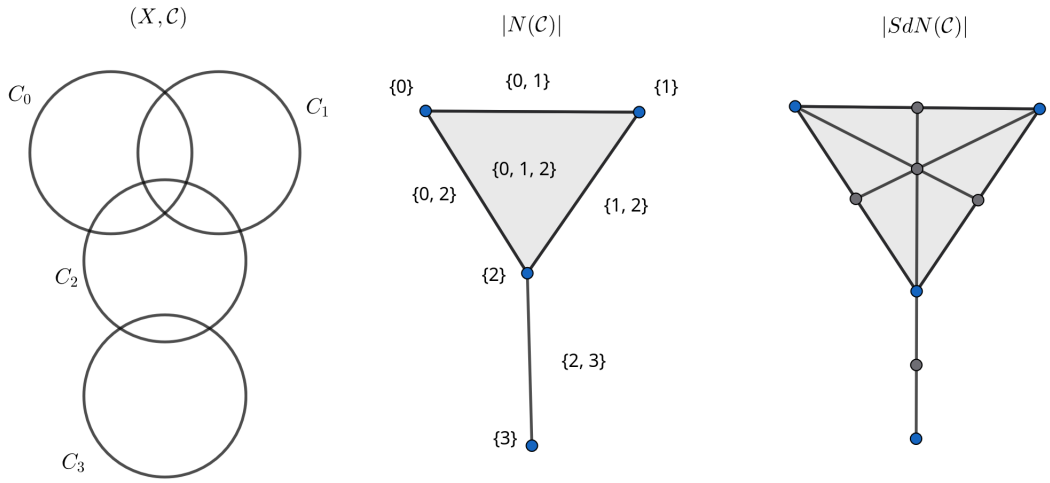


Figure 1: Example topological spaces

3.1 Γ function

Let $C_J := \bigcap_{j \in J} C_j$ for every $J \subseteq I$. From this point forward, we will only consider sets J for which C_J is not empty. There is a geometric vertex v_J in $|SdN(\mathcal{C})|$ that is the barycentric center of the simplex corresponding to J in $|N(\mathcal{C})|$.

To construct Γ function:

1. For each geometric vertex v_J in $|SdN(\mathcal{C})|$, we arbitrarily choose a point $x_J \in C_J$ and define $\Gamma(v_J) = x_J$.
2. The map is then extended to the entire complex by requiring it to be affine linear on each simplex.

For a geometric simplex σ with vertices $\{v_{J_0}, v_{J_1}, \ldots, v_{J_k}\}$, and $v \in \sigma$ we can write $v = \sum_{i=0}^{k} \lambda_i v_{J_i}$, where $\sum_{i=0}^{k} \lambda_i = 1$ and $\lambda_i \geq 0$. The function Γ is defined by:

$$\Gamma(v) = \sum_{i=0}^{k} \lambda_i \Gamma(v_{J_i}) = \sum_{i=0}^{k} \lambda_i x_{J_i}.$$

3.2 Φ function

When forgetting simplicial structure in geometric realization of $N(\mathcal{C})$ and $SdN(\mathcal{C})$ we get the same topological spaces. Because of this constructing map Φ from X to $|N(\mathcal{C})|$ will be the same as a map from X to $|SdN(\mathcal{C})|$.

3.2.1 Open cover $\mathcal{U}$

First we construct an open cover $\mathcal{U}$ of X, indexed by I such that for every i in I, $C_i \subseteq U_i$, and $N(\mathcal{U}) = N(\mathcal{C})$.

We will set $U_i^\epsilon = \{x \in X | d(x, C_i) < \epsilon\}$. It is trivial that with positive ϵ, $N(\mathcal{C}) \subseteq N(\mathcal{U}^\epsilon = \{U_i^\epsilon\}_{i \in I})$. What is left to show is that choosing properly small ϵ will yield $N(\mathcal{U}^\epsilon)) \subseteq N(\mathcal{C})$.

Let $J \subseteq I$ and let

$$far(J) = min_{x \in X}(max_{j \in J} d(x, C_j)))$$

Minimum in far exists because sets in $\mathcal{C}$ are compact. (It would be also true if the sets were only closed but the proof is harder.)

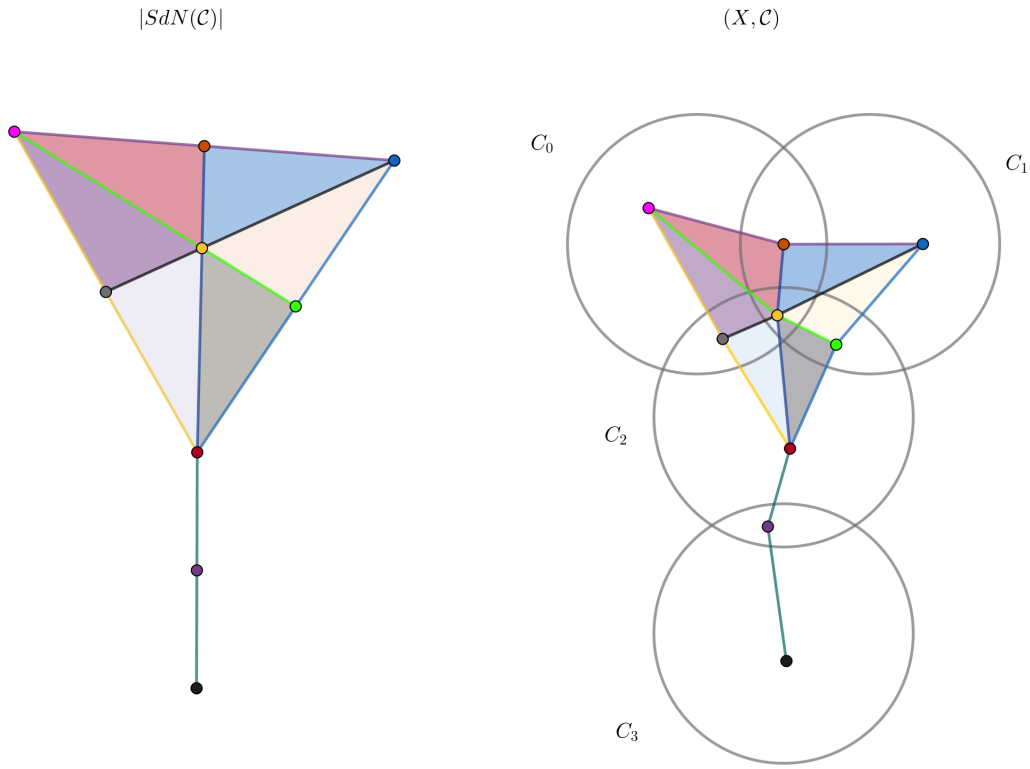


Figure 2: Gamma function

Now it suffices to set ϵ as the minimum of values of $far(J)$ over all $J \subseteq I$ such that $\bigcap_{i \in J} C_i$ is empty. (If there are no such subsets, then any ϵ is fine.)

If with such ϵ any new simplex corresponding to subset $J \subseteq I$ would be created in $N(\mathcal{U}^\epsilon)$ it would imply existence of a point x, such that $max_{j \in J} d(x, C_j) < \epsilon$ which is a contradiction. It's worth to note that $\epsilon > 0$ so the new cover will actually contain new points and be open.

3.2.2 Constructing Φ

We will construct partition of unity subordinate to this open cover $\mathcal{U} = \{U_i^\epsilon \cap X\}$, which is family of continuous functions $\{\phi_i\}_{i \in I}$ such that with $x \in X$:

$$\sum_{i \in I} \phi_i(x) = 1$$

$$\phi_i(x) \in [0, 1]$$

and ϕ_i is 1 on C_i and 0 outside of U_i.

To get those functions ϕ_i we can for example set:

$$\psi_i = \frac{d(x, \mathbb{R}^d \backslash U_i)}{d(x, C_i) + d(x, \mathbb{R}^d \backslash U_i)}$$

And normalize them $phi_i = \frac{\psi_i}{\sum_{j \in I} \psi_i}$

Now we can set $\Phi(x) = \sum_{i \in I} \phi(x) |v_i|$ where $|v_i|$ is geometric realization of vertex in $N(\mathcal{C})$ corresponding to $\{i\}$.

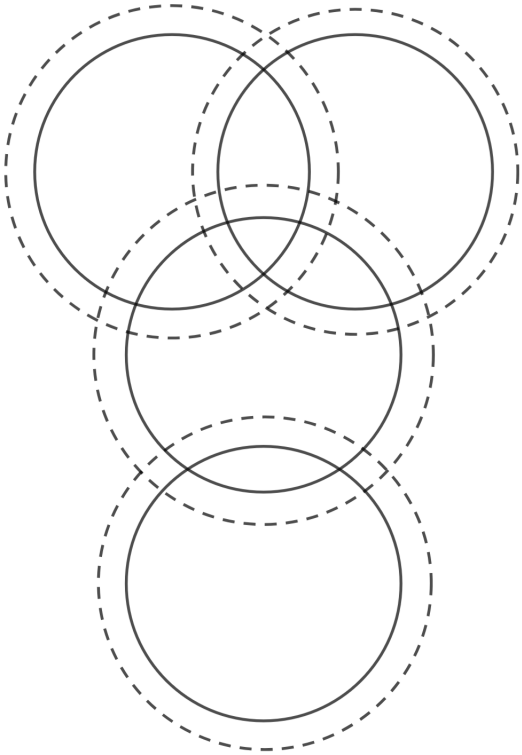
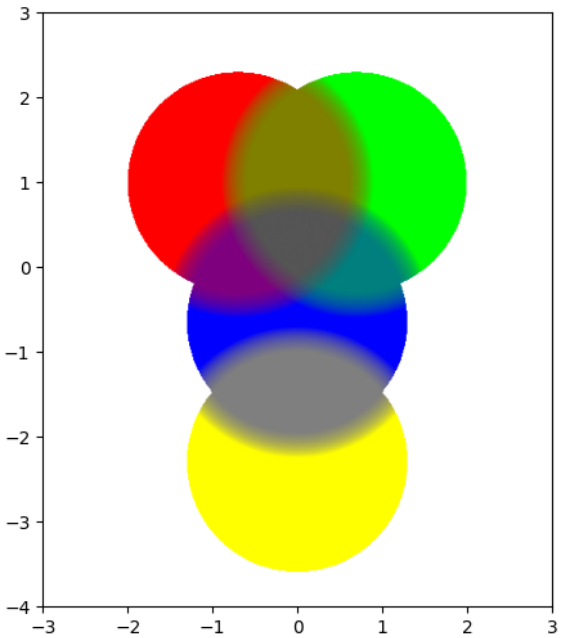
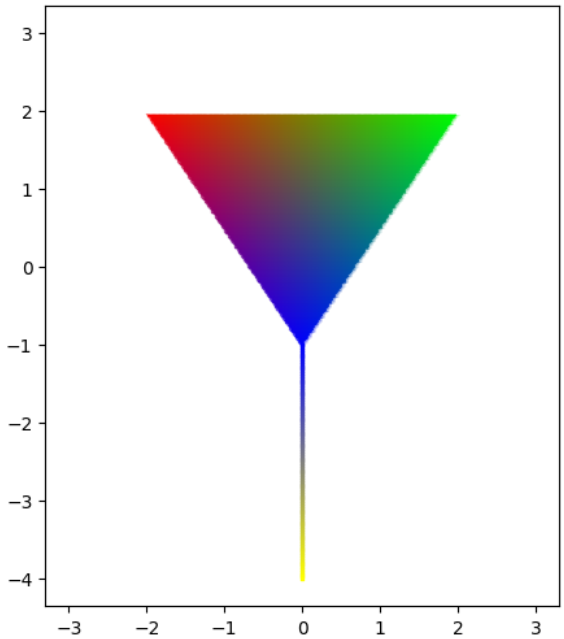


Figure 3: covers: C and U



(a) Phi on X



(b) X mapped onto N(X)

Figure 4: Phi function

3.3 Barycentric coordinates and closed barycentric stars

Let us set up some notation: Let K be a simplicial complex and w be it's vertex. We can define closed barycentric star of w as:

$$bst(w) = |\{\sigma \in SdK | \sigma \cup \{v\} \in SdK\}|$$

Simplices $\alpha \in SdK$ will be written as

$$\alpha = \{\tau_0 \subseteq \tau_1 \subseteq \cdots \subseteq \tau_k\} := \{z(\tau_0), z(\tau_1) \dots z(\tau_k)\}$$

where τ_i are simplices in K and $z(\tau_i) \in VerSdK$ is barycenter of τ_i. Additionaly for $x \in |K|$ let us write $b_w(x)$ as the barycentric coordinate of x with respect to vertex w, and for $\sigma \in K$ $s(\sigma) \in |K|$ as the barycenter of $|\sigma|$. I will assume $s(\sigma) = |z(\sigma)|$

Now we will want to prove some theorems needed for the final part.

Theorem 1: Each finite intersection of closed barycentric stars in K is contractible.

To see this we will need:

Lemma 1: Let $\sigma \in K$ and $L = \{\{\tau_0 \subseteq \tau_1 \subseteq \cdots \subseteq \tau_k\} \mid \sigma \subseteq \tau_0\} \subseteq SdK$, then:

$$\bigcap_{v \in \sigma} bst(v) = |L|$$

Proof of **Lemma 1**: if $\alpha \in L$ then for every $v \in \sigma$: $\alpha \cup \{v\}$ is a simplex in SdK. This means that $|\alpha| \in bst(v)$. It folows that $|L| \subseteq \bigcap_{v \in \sigma} bst(v)$.

Proof of **Theorem 1**: Let us take vertices $v_0, v_1 \dots v_m \in VerK$. If $S = \{v_0, \dots v_m\}$ is not a simplex in K, then for any $i \in [m]$ and $x \in bst(v_i)$ there exist a simplex $v_i \in \sigma \subset S$ and a vertex $v_j \in S$ that is not in σ. Then x is not contained in $bst(v_j)$, which means intersection is empty. If S is a simplex in K then by **Lemma 1** every geometric simplex in $\bigcap_{i \in [m]} bst(v_i)$ has a coface containing $s(S)$. This means that $\bigcap_{i \in [m]} bst(v_i)$ is star-shaped with respect to $s(S)$.

Now if $\alpha = \{\tau_0 \subseteq \tau_1 \subseteq \cdots \subseteq \tau_k\} \subseteq \bigcap_{v \in \sigma} bst(v)$ then for every $v \in \sigma$, there exist a simplex $\{v\} \cup \alpha$ in SdK, which means v is contained in every τ_j. From here follows that $\sigma \subseteq \tau_j$ so $\alpha \in |L|$ which proves inclusion the other way.

Theorem 2: Other representation of $bst(v)$: point $x \in |K|$ fulfills:

$$b_v(x) \geq b_w(x) \text{ for all } w \in VerK$$

if and only if $x \in bst(v)$.

To see this we will need two lemmas:

Lemma 2.1: Let us write $x \in |K|$ in sorted barycentric coordinates of K:

$$\sum_{i=0}^{m} \alpha_i |w_i|$$

where

$$\alpha_0 \geq \alpha_1 \geq \cdots \geq \alpha_m$$

Let simplices $\tau_i = \{w_0, \dots_i\}$, then $x \in |\{s(\tau_0), s(\tau_1), \dots\ s(\tau_m)\}|$.

Proof:

$$\sum_{i=0}^{m} \alpha_i |w_i| = \sum_{i=0}^{m} (\sum_{j=0}^{i} |w_i|)(\alpha_j - \alpha_{j+1}) = \sum_{i=0}^{m-1} s(\tau_i)(j+1)(\alpha_j - \alpha_{j+1}) + (m+1)\alpha_m$$

In the last equality $s(\tau_i) = \frac{\sum_{j=0}^{i} |w_i|}{j+1}$. All of the barycentric coordinates $\beta_i = (j+1)(\alpha_j - \alpha_{j+1}) \geq 0$, and $\beta_m = (m+1)\alpha_m$.

Lemma 2.2: Let $x \in |K|$ be written in barycentric coordinates of SdK:

$$x = \sum_{i=0}^{m} \beta_i s(\tau_i)$$

for simplices $\tau_0 \subseteq \tau_1 \subseteq \cdots \subseteq \tau_m$, where $\tau_i = |\{w_0, w_1, \dots_m\}|$ for $w_i \in VerK$. When writing x in barycentric coordinates of K where α_i is corresponding to vertex w_i we have that:

$$\alpha_i = \sum_{j=1}^{m} \frac{1}{j+1} \beta_j$$

Proof: using **Lemma 2.1** we have:

$$\alpha_j = \alpha_{j+1} + \frac{1}{j+1}\beta_j = \alpha_{j+2} + \frac{\beta_{j+1}}{j+2} + \frac{\beta_j}{j+1}$$

and so on up to α_m.

Proof of **Theorem 2**: Implication (=>): Let $w_0, w_1, \dots w_m$ be vertices of K such that $b_{w_0}(x) \geq w_1 \geq \dots w_m \geq 0$ and $v = w_0$. Then by **Lemma 2.1** x is in $|\{s(\tau_0), s(\tau_1), \dots\ s(\tau_m)\}|$ with $\tau_0 = v$, which is part of $bst(v)$ by definition.

Implication(=>): From **Lemma 2.2** we know that when $\sigma \in K$ and $\tau \in SdK$, $|\tau| \subseteq |\sigma|$, then barycentric coordinate corresponding to vertex $v \in VerK$ that is a part of τ as well as of σ will be maximal.

3.4 Γ and Φ give us morphisms of covered spaces

Fact: (Φ, id_I) is a morphism of covered spaces:

$$(X, (C_i)_{i\in I}) \to (|SdN(\mathcal{C})|, (bst(v_i))_{i\in I})$$

Proof: If $x \in C_i$ then $\phi_i(x) = 1$ and so it is maximal among barycentric coordinates $(\phi_j)_{j\in I}$ so from **Theorem 2** we know that $\Phi(x)\ inbst(v_i)$.

Fact: (Γ, id_I) is a morphism of covered spaces

$$(SdN(\mathcal{C}), (bst(v_i))_{i\in I}) \to (X, (C_i)_{i\in I})$$

Proof: by definition Γ sends vertices of simplices in $bst(v_i)$ to subsets of C_i. C_i are convex and Γ is affine linear so everything works.

3.5 Constructing homotopies

3.5.1 $\Gamma \circ \Phi$

It follows that $(\Gamma\circ\Phi, id_I)$ is a morphism of covered spaces. Because $\Gamma\circ\Phi(C_i) \subseteq C_i$ and from convexity of C_i every two points in C_i can be connected by a straight line we can construct a homotopy: $H(x,t) = (1-t)x + t(\Gamma\circ\Phi)(x)$. So $\Gamma \circ \Phi$ is homotopic to identity.

3.5.2 $\Phi \circ \Gamma$

Here we will need a lemma (which apparently is pretty standard argument in homotopy theory):

Lemma 3: A continuous map g from a sphere S^n to contractible space W can be extended to a continuous map G from a ball B^{n+1} into W.

Proof: Contraction of W induces a homotopy $H : S^n \times [0,1] \to W$ between G and a constant map. We also have surjection $sur : S^n \times [0,1] \to B^{n+1}$, $sur(x,r) = x * r$. G will be now determined by requiring that $G \circ sur = H$.

Lets remind that $\mathcal{C}$ - closed convex cover, I - indexing set of $\mathcal{C}$, $(v_i)_{i\in I}$ vertices of $N(\mathcal{C})$. Now lets show that $f := \Phi \circ \Gamma$ is homotopic to identity on $K := SdN(\mathcal{C})$. We will inductively construct homotopies $H^m : |sk_mK| \times [0,1] \to |K|$ (where sk_mK is m-skeleton of K) between restrictions $id|_{sk_mK}$ and $f|_{sk_mK}$ such that for all $i \in I$ homotopy $H^m((bst(v_i) \cap |sk_mK|) \times [0,1]) \subseteq bst(v_i)$. When $m = dimK$ then H^m will be desired homotopy.

For the base case when $m = 0$ first take any vertex p of K. Then let $J \subseteq I$ be those indices that for $j \in J$ satisfy $p \in bst(v_j)$. Function f is carried by identity so both p and $f(p)$ must lie in $W := \bigcap_{j\in J} bst(v_j)$. And because W is contractible from **Theorem 1** we can connect p and $f(p)$ by a path lying in W which will give our homotopy H^0.

Now for the induction step from (m-1) to m, let H^{m-1} satisfy the induction hypothesis and σ be m-simplex in sk_mK. Similarly to base case let $J \subseteq I$ be those indeces that satisfy $\forall_{j\in J}\sigma \subseteq bst(v_j)$. Then both σ and $f(\sigma)$ must lie in $W := \bigcap_{j\in J} bst(v_j)$. From the induction hypothesis we also have $H^{m-1}(|\partial\sigma| \times [0,1]) \subseteq W$. Now let $L = (|\partial\sigma| \times [0,1]) \cup (|\sigma| \times 0, 1)$. Our space L is homeomorphic to a sphere S^m and $|\sigma| \times [0,1]$ is homeomorphic to a ball B^{m+1}. Now we can use **Lemma 3**. Function g from $L \cong S^m$ to W is given by homotopy $H^{m-1}|_{|\partial\sigma|\times[0,1]}$ on $|\partial\sigma| \times [0,1]$, by $f|_{\sigma|}$ on $|\sigma| \times 1$ and by $id|_{\sigma|}$ on $|\sigma| \times 0$. Extension described in lemma will be a homotopy $H^m|_{\sigma\times[0,1]}$ We can repeat this for every m-simplex σ and get the full homotopy H^m.

4 Where Nerve Theorem is used

Nerve Theorem as I have stated here, is for example used in:

- **Čech complex:** We have a point clound in Euclidean space and for a given r, we construct cover by taking an r-ball around each point. Čech complex is the nerve of the r-balls. Nerve Theorem guarantees that the Čech complex is homotopy equivalent to the union of balls.
- **Delaunay–Čech complex:** An in the Čech complex, but the cover is taken as an r-ball around a point, intersected with Voronoi cell of this point. Both are closed and r-ball is compact so the conditions are met.
- **Mapper algorithm:** Constructs cover of data from pre-images of filter functions, and then build its nerve. The Nerve Theorem ensures that the Mapper complex is homotopy equivalent to the underlying space.